THE CHINA SHOCK AND COMMUTING ZONE CHARACTERISTICS

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I: Introduction

One of the most important findings in the last 20 years of research on globalization and inequality is that classical trade theory's assumption of "perfect labor mobility" is not strongly violated by real world evidence. The genesis of this comes from Autor, Dorn, and Hanson's 2013 paper on distributional consequences of increased manufacturing trade exposure to China.

In the following decade, hundreds of studies have used this framework to examine why trade affects some places more than others, contradicting years of established economic thought. The "China shock", a period in American trade history following China's ascension to the World Trade Organization (WTO), provides a useful setting to study trade and inequality. Beyond the trade literature, there is also significant work in the field of labor economics which studies how and why people move to new jobs. This paper seeks to study the impact of local labor market characteristics on worker mobility, building upon established work on the effects of demographics and industrial composition (eg. Autor et al., 2021; Greenland et al. 2019).

Using data from Autor, Dorn, and Hanson and a metric for the concentration of large labor markets nearby, I study migration responses to the China shock over the period of 1990-2007 at the commuting zone level to see if and why some regions react differently. This paper seeks to add to the understanding of differential labor market responses to trade shocks and help further explain the intensity and longevity of its negative effects.

The paper is organized as follows. Section 2 provides an overview of relevant literature. Section 3 provides some theoretical background for the model of labor mobility. Section 4 discusses data collection and definitions. Section 5 provides estimates of regional effects and analysis of results. Lastly, section 6 summarizes and concludes the paper.

II: Literature Review

Numerous studies in the last twenty years have been published assessing, quantifying, and interpreting the distributional consequences of trade liberalization. Recently, following Autor Dorn, and Hanson's 2013 analysis of increased trade with China and its effects on the manufacturing population, many researchers have used the "China Shock" period of approximately 2000-2010 to study trade and unequal effects on US regions. The most significant finding was the relative immobility of labor, as opposed to classical theory's assumptions. More modern models of trade like Artuc et al (2010) suggest a slow reallocation of workers following trade liberalization. In general, McLaren (2022) finds that research shows that switching industries and occupations in response to trade shocks is costly, and that adjustments vary substantially by gender and age.

Autor, Dorn, and Hanson's 2016 paper emphasizes the need for research on the topic to study heterogeneity in responses to demand shocks. Pierce and Schott (2016) use a different identification strategy to study employment and migration effects of the increase in Chinese imports and find similar results. Greenland, Lopresti and McHenry (2019) use that identification strategy to look at the responses of different groups, finding migration responses to be strongest among younger and foreign-born workers, especially men. Autor Dorn and Hanson's 2023 paper finds similar results for the characteristics of workers that increase the likelihood of migrating in response to trade and find that a commuting zone's initial share of college-educated workers and level of industrial specification are important determinants to the severity of the impact. It also finds significant long-term effects following the period of the rapid trade increase with China.

Faber Sarto and Tabellini (2022) compare the China shock's effects to that of the increased use of robots, and find that the trade shock did not cause substantial migration across commuting zones. Bartik (2018) focuses on moving costs, and the factors that would impede an individual from moving. Galle et al (2023) finds that the negative effects of the China Shock are especially concentrated in the Midwest and Inland East regions of the United States.

Papers like Hyman (2018) and Chan (2019) study policy and local characteristics, and the former finds a small impact of Trade Adjustment Assistance on worker outcomes, while the latter finds that some policies intended to protect workers can exacerbate the negative impacts of a trade shock.

This phenomenon has also been studied in other countries, like Ashournia (2018) in Denmark and Dix Carneiro and Kovak (2017) in Brazil. The latter finds increasing negative effects of trade on labor market outcomes over time.

There is also a body of literature looking at migration responses in the United States in response to the Great Recession. Yagan (2019) finds that more severe local exposure to the negative demand shock caused working-age Americans to be less likely to be employed, even with a local recovery in unemployment rates. Cardena and Kovak (2016) found significant mobility responses from low-skilled Mexican-born workers in the United States due to the recession, and their work provides evidence that immigrants respond to labor market conditions by moving. Monras (2015) finds very little overall internal migration change in response. Foote et al. (2015) study all potential channels for leaving the labor force and find that migration in response to demand shocks was significant. Finally, Monte et al. (2018) show that the "commuting openness" of a county or commuting zone significantly impacts how negative demand shocks impact it.

III: Theoretical Approach

Following Autor et al. (2013), I use the "gravity" structure model mapping changes in trade quantities to labor market outcomes, as opposed to trade prices. Solving the model gives the following equations for wage and employment in traded and nontraded goods:

$$(1)\ \ \widehat{W_i} = \sum_j c_{ij} \frac{L_{ij}}{L_{Ni}} \left[\theta_{ijC} \widehat{E_{Cj}} - \sum_k \theta_{ijk} \varphi_{Cjk} \widehat{A_{Cj}} \right]$$

$$(2)\ \ \widehat{L_{Ti}} = \rho_i \sum_j c_{ij} \frac{L_{ij}}{L_{Ti}} \left[\theta_{ijC} \widehat{E_{Cj}} - \sum_k \theta_{ijk} \varphi_{Cjk} \widehat{A_{Cj}} \right]$$

$$(3)\ \ \widehat{L_{Ni}} = \rho_i \sum_j c_{ij} \frac{L_{ij}}{L_{Ni}} \left[-\theta_{ijC} \widehat{E_{Cj}} + \sum_k \theta_{ijk} \varphi_{Cjk} \widehat{A_{Cj}} \right]$$

By making the assumptions on ρ_{ij} and c_{ij} in Autor et al. (2013) and focusing solely on the Chinese imports channel, we get the following shift-share measure for the change in imports per worker:

$$(4)\ \ \Delta \mathrm{IPW}_{uit} = \sum_j \frac{L_{ijt}}{L_{ujt}} \frac{\Delta M_{ucjt}}{L_{it}}$$

There is an issue with correlation for the above equation, which is that the possible relationship between imports from China and unobserved changes to demand for certain products in the US. Thus, we also use their instrument for the change in imports per worker, which uses data on products from China imported to non-US countries:

$$(5)\ \ \Delta \text{IPW}_{oit} = \sum_{j} \frac{L_{ijt-1}}{L_{ujt-1}} \frac{\Delta M_{ocjt}}{L_{it-1}}$$

This paper also uses a measure of how much a state's population is accounted for by a commuting zone. Using commuting zones, developed by Tolbert and Sizer (1996), as opposed to counties or cities, is the standard practice in China Shock literature. A commuting zone is a region in which many people commute to work *within* a commuting zone, and few people commute to work *outside* of a commuting zone. This helps us to get an accurate idea of local labor markets. My measure will represent how many nearby employment opportunities a worker affected by the trade shock will have. The idea is that a worker who lives an hour from a larger commuting zone would be more likely to move there for work than someone who would have to move potentially 10 hours away. The "Commuting Zone State Share" measure for commuting zone *i* and state *s* is:

$$(6)\ \ \text{CZS}S_{is} = \frac{L_i}{L_s}$$

The measure that will be used in the empirical portion of this paper will be the "Rest of State" measure, equal to:

$$(7)\ \ ROS_{is} = 1 - \frac{L_i}{L_s}$$

This will capture the portion of a state's population that lives outside of a particular commuting zone and should be a measure of how close you are to another reasonably sized commuting zone.

IV: Data

Data for this paper comes from publicly available data from Autor et al., obtained from OpenICPSR. Their data on US imports comes from the UN Comrade Database, and data on commuting zones comes from Tolbert and Sizer (1996). They get industry employment data from County Business Patterns data, and then convert it to the commuting zone level. Finally, they get demographic controls from Census Integrated Public Use Micro Samples, and then map this to commuting zones.

V: Empirical Approach

The following regression equation will be the base specification for this paper:

$$(8)\ \Delta L_{it} = \beta_0 + \beta_1 \Delta IPW_{uit} + \beta_2 \mathrm{RO}S_{is} + \beta_3 \mathrm{RO}S_{is}\Delta IPW_{uit} + \beta_4 ForeignBorn + \beta_5 L_{it} + \Theta_t + \Theta_s + \varepsilon_{it}$$

The dependent variable is the change in population in a commuting zone over a given time. The first beta represents the change in imports per worker, the second is the effect of the "Rest of State" variable, and then the third is the variable of interest, the interaction between the rest of state measure and the shift-share measure for change in imports per worker. The foreign-born population in the commuting zone and the population size are controls suggested by the literature. Lastly, the Θ_t and Θ_s terms are time and state fixed effects respectively.

Next, I will switch from the potentially endogenous ΔIPW_{uit} to the instrument, $\Delta \mathrm{IPW}_{oit}$, which uses lagged information on Chinese exports to other high-income countries. The second specification I run is the following:

$$(9)\ \Delta L_{it} = \beta_0 + \beta_1 \Delta \mathrm{IPW}_{oit} + \beta_2 ROS_{is} + \beta_3 ROS_{is} \Delta \mathrm{IPW}_{oit} + \beta_4 ForeignBorn + \beta_5 L_{it} + \Theta_t + \Theta_s + \varepsilon_{it}$$

Finally, I also want to study this problem using regional effects – are some regions more filled with dense commuting zones with lots of employment opportunities, and does this impact how they are effected by trade shocks? Of course, this is not a causal story; the purpose of these regressions is to provide a starting point for future work in broad regional variation in trade or demand shock effects. The specifications I run for this are the following:

$$(10)\ \Delta L_{it} = \beta_0 + \beta_1 \Delta IPW_{uit} + \beta_2 ForeignBorn_i + \beta_3 L_{it} + \beta_4 Region_i \Delta IPW_{uit} + \Theta_t + \Theta_s + \varepsilon_{it}$$

$$(11)\ \Delta L_{it} = \beta_0 + \beta_1 \Delta IPW_{oit} + \beta_2 ForeignBorn_i + \beta_3 L_{it} + \beta_4 Region_i \Delta IPW_{oit} + \Theta_t + \Theta_s + \varepsilon_{it}$$

VI: Results

Table 1 shows summary statistics for the "Rest of State" measure. Its mean is 0.94 and its median is 0.98, which indicates that the mean and median US commuting zones occupy very small fractions of a state's population. It has a relatively large standard deviation, approximately 13 percentage points, which is to be expected as some major cities occupy very large percentages of state populations. Figure 1 is a frequency histogram that shows the distribution of the "Rest of State" variable. It shows that most commuting zones are very small fractions of a state's population, which makes intuitive sense.

Table 2 reports regression results for the first specification, from equation (8). The coefficient of interest, for the interaction between "Rest of State" and ΔIPW_{uit}, is not significant, indicating that when just using OLS, there is no significant difference between living in a place with lots of large nearby commuting zones and a state with fewer options in terms of a migration response. However, the coefficient on "Rest of State" is significant, indicating that for reasons unrelated to trade or demand shocks, people may be leaving commuting zones with fewer local options more often than those with more. State dummy coefficients are suppressed in all regression results.

Table 3 shows regression results for the specification detailed in equation (9), using the instrument with imports to other high-income countries. The results in this table are largely similar to the results in table 2, suggesting that perhaps the “Rest of State” measure, and the thing that it attempts to capture, is an especially important determinant of labor mobility in response to trade shocks. However, the negative and significant coefficient on the non-interacted “Rest of State” term implies that these areas may be evolving differently population-wise and may be an area to look further into in future studies.

Table 4 includes regression results for the specification in equation (10), looking at different responses to the China shock across census-defined regions. The results show some difference in signs on the coefficients of the region interaction terms, but of course none of them reach the standard thresholds for statistical significance. In table 5, results are presented for equation (11), which uses the instrument for the change in imports per worker. Once again, there are differences in signs and magnitudes of the estimated coefficients on the region interaction terms, but none of them are statistically significant so we should not try to make conclusions based on them.

Table 1: Summary of "Rest of State" variable

```
. sum rest_of_state, d

                        rest_of_state
-------------------------------------------------------------
      Percentiles      Smallest
 1%     .3414865              0
 5%     .7320119              0
10%     .8418813              0       Obs               1,444
25%     .9412203              0       Sum of wgt.       1,444

50%     .9757896                      Mean            .933518
                        Largest       Std. dev.      .1270474
75%     .9899414       .9998227
90%     .9964332       .9998581       Variance        .016141
95%     .9980837       .9999098       Skewness      -4.083624
99%     .9994473       .9999282       Kurtosis       22.53799

.
```

Figure 1: Histogram for "Rest of State" variable

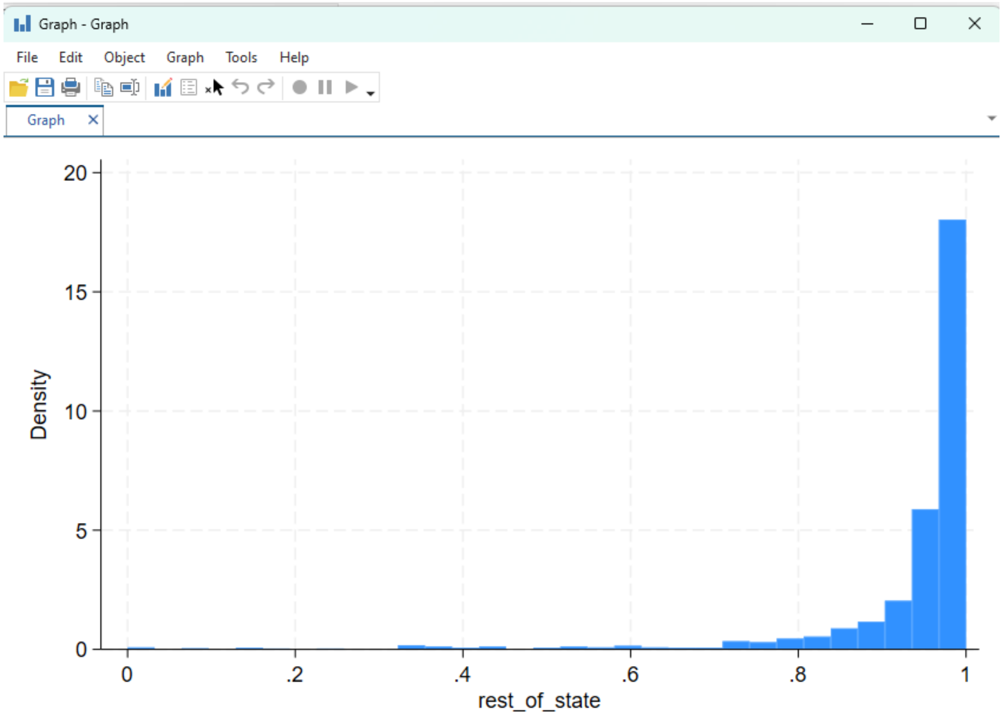

Table 2: Regression results 1

```
.
. reg lnchg_popworkage d_tradeusch_pw l_sh_popfborn l_popcount rest_of_state rest_of_statex_1 t2 i.statefip, robust

Linear regression                               Number of obs     =      1,444
                                                F(53, 1390)       =      18.31
                                                Prob > F          =     0.0000
                                                R-squared         =     0.3846
                                                Root MSE          =      8.317
```

lnchg_popworkage	Coefficient	Robust std. err.	t	P>|t|	[95% conf.	interval]
d_tradeusch_pw	.7311444	1.471303	0.50	0.619	-2.155069	3.617358
l_sh_popfborn	.1664891	.0740131	2.25	0.025	.0212998	.3116785
l_popcount	-1.23e-06	3.44e-07	-3.56	0.000	-1.90e-06	-5.50e-07
rest_of_state	-20.17653	4.763643	-4.24	0.000	-29.52124	-10.83182
rest_of_statex_1	-.6129708	1.504226	-0.41	0.684	-3.563769	2.337828
t2	-3.84815	.49061	-7.84	0.000	-4.810566	-2.885734

Table 3: Regression results 2

Linear regression						
				Number of obs	=	1,444
				F(53, 1390)	=	18.22
				Prob > F	=	0.0000
				R-squared	=	0.3849
				Root MSE	=	8.3152

lnchg_popworkage	Coefficient	Robust std. err.	t	P>|t|	[95% conf.	interval]
d_tradeotch_pw_lag	-.9584648	1.483531	-0.65	0.518	-3.868667	1.951737
l_sh_popfborn	.1677893	.0740255	2.27	0.024	.0225755	.3130032
l_popcount	-1.21e-06	3.52e-07	-3.45	0.001	-1.90e-06	-5.24e-07
rest_of_state	-23.02219	4.932232	-4.67	0.000	-32.69761	-13.34676
rest_of_statex_2	1.183749	1.516277	0.78	0.435	-1.790689	4.158187
t2	-3.836756	.5215652	-7.36	0.000	-4.859896	-2.813616

Table 4: Regression results 3

Linear regression						
				Number of obs	=	1,444
				F(59, 1384)	=	16.97
				Prob > F	=	0.0000
				R-squared	=	0.4101
				Root MSE	=	11.603

lnchg_pop~1634	Coefficient	Robust std. err.	t	P>|t|	[95% conf.	interval]
midatlx_1	-.779376	1.885356	-0.41	0.679	-4.47784	2.919088
encenx_1	-.4530426	1.482245	-0.31	0.760	-3.360732	2.454647
wncenx_1	-.0456314	1.492035	-0.03	0.976	-2.972526	2.881263
satlx_1	-.1830088	1.518279	-0.12	0.904	-3.161385	2.795367
escenx_1	-.0065891	1.477352	-0.00	0.996	-2.904681	2.891502
wscenx_1	.2822159	1.609452	0.18	0.861	-2.875013	3.439445
mountx_1	2.27886	1.955078	1.17	0.244	-1.556377	6.114097
pacifx_1	-2.006752	1.866729	-1.08	0.283	-5.668676	1.655172
d_tradeusch_pw	.0094646	1.471132	0.01	0.995	-2.876424	2.895354
l_sh_popfborn	-.1411805	.1163907	-1.21	0.225	-.3695018	.0871409
l_popcount	-1.22e-07	3.40e-07	-0.36	0.720	-7.90e-07	5.46e-07
t2	12.97442	.695422	18.66	0.000	11.61023	14.33862

Table 5: Regression results 4

Linear regression						
			Number of obs	=	1,444	
			F(59, 1384)	=	16.97	
			Prob > F	=	0.0000	
			R-squared	=	0.4101	
			Root MSE	=	11.603	

lnchg_pop~1634	Coefficient	Robust std. err.	t	P>|t|	[95% conf.	interval]
midatlx_1	-.779376	1.885356	-0.41	0.679	-4.47784	2.919088
encenx_1	-.4530426	1.482245	-0.31	0.760	-3.360732	2.454647
wncenx_1	-.0456314	1.492035	-0.03	0.976	-2.972526	2.881263
satlx_1	-.1830088	1.518279	-0.12	0.904	-3.161385	2.795367
escenx_1	-.0065891	1.477352	-0.00	0.996	-2.904681	2.891502
wscenx_1	.2822159	1.609452	0.18	0.861	-2.875013	3.439445
mountx_1	2.27886	1.955078	1.17	0.244	-1.556377	6.114097
pacifx_1	-2.006752	1.866729	-1.08	0.283	-5.668676	1.655172
d_tradeusch_pw	.0094646	1.471132	0.01	0.995	-2.876424	2.895354
l_sh_popfborn	-.1411805	.1163907	-1.21	0.225	-.3695018	.0871409
l_popcount	-1.22e-07	3.40e-07	-0.36	0.720	-7.90e-07	5.46e-07
t2	12.97442	.695422	18.66	0.000	11.61023	14.33862

VII: Conclusion

This paper studied whether the position in a network of local commuting zones had an impact on the migration response to trade shocks, using the 2000-2010 China Shock. It used data and a theoretical framework from Autor et. al. (2013), and introduced a metric for a commuting zone's nearness to larger, presumably more industrially diverse commuting zones. My results find no significant impact of this "rest of state" effect on migration in response to demand shocks, but hopefully future research will develop new ways to measure this and pin down a way to isolate and identify these effects more closely. This is an important problem to study, because understanding it better should lead to an improved ability to mitigate future trade shocks. It would allow economists to better predict which areas would be most affected and through which channels the effects tend to come.

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